



Exercise 1. Stabilization of the Furuta pendulum



Background

The Furuta pendulum is a classical example of an underactuated mechanical system. It consists of a rotating horizontal arm with a pendulum attached at its end. The pendulum is actuated indirectly through a torque applied at the base of the rotating arm.

A nonlinear model of the system can be derived using the Euler-Lagrange equations. In this exercise, you will work with a simplified linearized and discretized model of the Furuta pendulum around the upright equilibrium.

The pendulum angle is denoted by θ , where

$$\theta = 0$$

corresponds to the upright position of the pendulum. The control objective is to make the pendulum converge to this upright equilibrium.

The state vector used in this exercise is

$$x_k = \begin{bmatrix} \theta_k \\ \dot{\theta}_k \\ \dot{\varphi}_k \end{bmatrix} = \begin{bmatrix} \text{pendulum angle}_k \\ \text{pendulum angular velocity}_k \\ \text{base angular velocity}_k \end{bmatrix}.$$

The input u_k is the torque applied at the base. The base angle itself is not included in the state vector, since it is not needed for the stabilization task considered here.

The sampling time is

$$T_s = 0.005 \text{ s.}$$

A discrete-time model of the pendulum behavior near the upright equilibrium point is

$$x_{k+1} = \underbrace{\begin{bmatrix} 1.0026 & 0.0050 & 0 \\ 1.0399 & 1.0026 & 0 \\ -0.0675 & -0.0002 & 1 \end{bmatrix}}_A x_k + \underbrace{\begin{bmatrix} -0.0843 \\ -33.7508 \\ 39.2131 \end{bmatrix}}_B u_k$$

The output is assumed to be the full state, i.e., $y_k = x_k$.

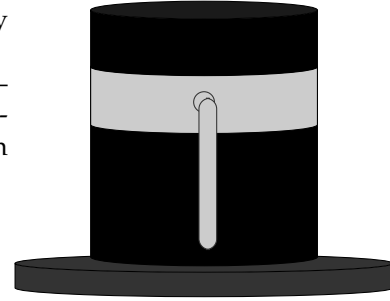
Objective

Your task is to design a discrete-time control law that uses the measured state x_k to compute the input u_k , such that the origin

$$x = 0$$

is an asymptotically stable equilibrium of the controlled system.

In physical terms, this means that the pendulum should return to the upright position with zero angular velocity and zero base velocity, for sufficiently small initial deviations from the upright equilibrium.



Tasks

1. Open-loop analysis

First consider the system without control input, i.e.

$$u_k = 0.$$

1. Determine the eigenvalues of A .
2. Based on these eigenvalues, decide whether the origin is stable, asymptotically stable, or unstable.
3. Explain the physical meaning of your conclusion.

2. Controllability

Investigate whether the input u_k can influence all relevant modes of the linearized system. Compute the reachability matrix

$$\mathcal{R} = [B \quad AB \quad A^2B].$$

1. Determine the rank of $\mathcal{R}$.
2. Explain what your result implies for the possibility of stabilizing the system.

3. Controller construction

Construct a control law of the form

$$u_k = \mu(x_k),$$

where $\mu(\cdot)$ is a function of the measured state.

The closed-loop system should satisfy

$$\theta_k \rightarrow 0, \dot{\theta}_k \rightarrow 0 \quad \text{as} \quad k \rightarrow \infty$$

for small initial conditions near the origin.

Your design should result in closed-loop dynamics of the form

$$x_{k+1} = A_{\text{cl}} x_k,$$

where all eigenvalues of A_{cl} lie strictly inside the unit circle. You may choose your preferred method for selecting the controller parameters.

4. Closed-loop verification

For your chosen controller:

1. Write down the closed-loop system matrix A_{cl} .
2. Compute the eigenvalues of A_{cl} .
3. Verify that all closed-loop eigenvalues lie strictly inside the unit circle.
4. Explain why this implies local asymptotic stability of the upright equilibrium for the linearized model.

5. Simulation

Simulate the closed-loop system for at least three different initial conditions close to the origin. For example, you may use initial conditions of the form

$$x_0 = \begin{bmatrix} \theta_0 \\ 0 \\ 0 \end{bmatrix},$$

with different small values of θ_0 .

For each simulation, plot:

- the pendulum angle θ_k ,
- the pendulum angular velocity $\dot{\theta}_k$,
- the base angular velocity $\dot{\varphi}_k$,
- the input torque u_k .

Use the sampling time $T_s = 0.005$ s when producing time-domain plots.

6. Discussion

Discuss the behavior of your controlled system. Your discussion should include:

1. how quickly the states converge to zero,
2. how sensitive the response is to the chosen controller parameters,
3. whether the same controller should be expected to work for large initial pendulum angles.

Instructions

Submit a ZIP file that includes at least a PDF and a code folder. The submission should contain:

- the open-loop eigenvalue analysis,
- the reachability calculation,
- a description of your controller design,
- the controller equation $u_k = \dots$,
- the closed-loop system matrix,
- the closed-loop eigenvalues,
- simulation plots,
- a short discussion of the results,
- the code used for your computations and simulations.